VAIBHAV PANCHAL

AI Engineer | LLM Applications and Agents

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PROFESSIONAL SUMMARY

AI/ML Developer at TestGrid Labs who builds and tests AI for CoTester, TestGrid's AI testing agent. Replaced a third-party LLM API with an in-house vision model, accelerated image-based element detection by ~85% and validates new agent releases. Architects LangGraph multi-agent and RAG systems with guardrails and evaluation.

TECHNICAL SKILLS

LLMs & Agents:	LangGraph, LangChain, Multi-Agent Systems, Tool Calling, RAG, NLP, Prompt Engineering, MCP
Models & Evaluation:	OpenAI API, Groq, Ollama, VLMs, HuggingFace Transformers, RAGAS, NeMo Guardrails, Langfuse
Backend & DevOps:	Python, FastAPI, REST APIs, Docker, AWS (EC2), Nginx, GitHub Actions CI/CD, Pytest
ML & Data:	PyTorch, TensorFlow, Scikit-learn, YOLO, OpenCV, ONNX, OCR, MongoDB, Pinecone, SQL
Frontend & Tools:	JavaScript, React, Streamlit, Git, Postman, Selenium Grid

PROFESSIONAL EXPERIENCE

Junior AI/ML Developer — TestGrid Labs Inc.
Navsari, Gujarat | May 2026 – Present

- Shipped an in-house vision pipeline (VLM, ONNX, two-pass OCR) on FastAPI, replacing a third-party LLM API.
- Optimized an image-based element detector, cutting processing time by ~85% and resolving user-reported defects.
- Evaluated CoTester releases: researched agent evaluation frameworks, authored test-case collections for new agents, tested AI features via REST APIs and assessed local-model deployment.
- Developed backend tooling for a secure tunneling product: log replication, usage tracking and overload warnings.

Trainee Associate Software Developer — Techreale Pvt. Ltd.
Valsad, Gujarat | Jan 2024 – Apr 2024

- Delivered Famyll healthcare platform features (React.js, Node.js, MongoDB) and refactored async API integrations.

FEATURED ENGINEERING PROJECTS

MESH Support: Multi-Agent Customer Support Platform
Live Demo | GitHub | 2026

Tech Stack: Python, LangGraph, FastAPI, MongoDB, Groq GPT-OSS-120B, NeMo Guardrails, Langfuse, Pytest

- Architected a LangGraph Supervisor routing queries to Billing, Technical and Refund sub-agents with a loop guard.
- Implemented fail-closed NeMo guardrails: 100% precision/recall, 0% false positives on 35 messages, in two runs.
- Achieved 97.5% end-to-end accuracy (39/40) over two runs of a 20-case live-API suite, with server-side access control, Langfuse tracing, 23/23 live E2E tests and 137 offline tests.

RAG System with Evaluation Pipeline
Live Demo | GitHub | 2026

Tech Stack: Python, FastAPI, React, LangChain, Pinecone, BM25, RAGAS, Langfuse, MongoDB, Docker

- Engineered hybrid retrieval fusing Pinecone and BM25 via Reciprocal Rank Fusion, with keyword-only fallback.
- Built a 45-question golden set scoring RAGAS, retrieval hit rate/MRR and refusals; 72 offline tests run in CI.

ExamForge: AI Exam Question Generator
GitHub | 2026

Tech Stack: Python, Streamlit, OpenAI API, Ollama, ReportLab

- Automated exam-paper creation from syllabus PDFs using local (Ollama) or cloud (OpenAI) models.

DEEP LEARNING RESEARCH

Channel Attention for Oriented Object Detection in Remote Sensing
2025 (Under Review)

- Enhanced YOLOv11s-OBB with ECA, gaining +3.8% mAP50 and +4.3% recall on DIOR-R with no added parameters.

EDUCATION

Master of Engineering (M.E.), Software Engineering
CGPA: 8.21 / 10 | 2024 – 2026

L.D. College of Engineering (GTU), Ahmedabad

Bachelor of Engineering (B.E.), Computer Engineering
CGPA: 8.11 / 10 | 2020 – 2024

GDEC (GTU), Navsari

CERTIFICATIONS

- **Microsoft Learn:** Evaluate Generative AI Applications (Azure AI Foundry, Content Safety Tools) (2026)
- **Anthropic:** Model Context Protocol — Introduction & Advanced Topics (2026)